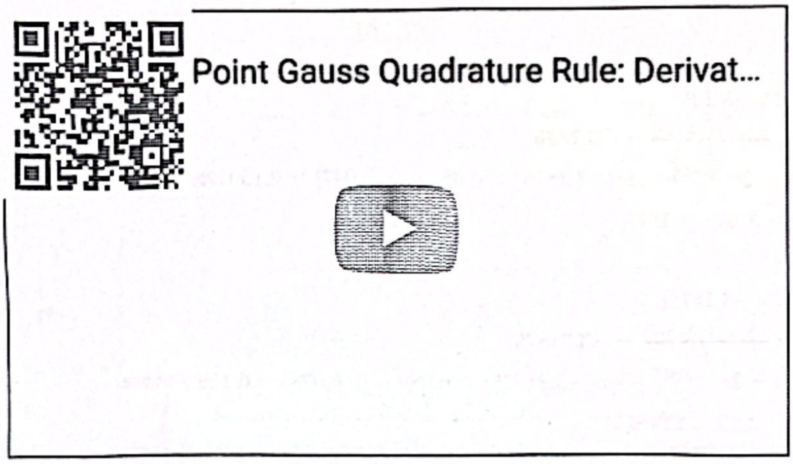

Lesson 2: Applications of Gauss Quadrature Rule

Learning Objectives

After successful completion of this lesson, you should be able to:

1) *use the Gauss quadrature rule to approximate definite integrals.*

Recap

In the previous lesson, you learned the theory of the Gauss quadrature rule. In this lesson, we apply the formulas of the rule to approximate definite integrals.

✓ Example [illegible]

The following integral is given

$$\int_{0.1}^{1.3} 5xe^{-2x}\, dx$$

a) Use the two-point Gauss quadrature rule to estimate the value of the integral.

b) Find the true error for part (a).

c) Find the absolute relative true error for part (a).

Solution

a) The two-point Gauss quadrature rule is given by

$$\int_a^b f(x)\, dx \approx c_1 f(x_1) + c_2 f(x_2)$$

where

$$c_1 = \frac{b-a}{2}$$

$$x_1 = \frac{b-a}{2}\left(-\frac{1}{\sqrt{3}}\right) + \frac{b+a}{2}$$

$$c_2 = \frac{b-a}{2}$$

$$x_2 = \frac{b-a}{2}\left(\frac{1}{\sqrt{3}}\right) + \frac{b+a}{2}$$

For this example

$$f(x) = 5xe^{-2x},\ a = 0.1,\ b = 1.3$$

So

4. **Iteration 4:**
   - Interval: $[-3.25, -3.125]$
   - Midpoint: $c = \frac{-3.25+(-3.125)}{2} = -3.1875$
   - $f(-3.1875) = 2e^{-3.1875} - \sin(-3.1875) \approx 0.083 - (-0.047) \approx 0.130$ (Positive)
   - New interval: $[-3.25, -3.1875]$
5. **Iteration 5:**
   - Interval: $[-3.25, -3.1875]$
   - Midpoint: $c = \frac{-3.25+(-3.1875)}{2} = -3.21875$
   - $f(-3.21875) = 2e^{-3.21875} - \sin(-3.21875) \approx 0.080 - (-0.078) \approx 0.158$ (Positive)
   - New interval: $[-3.25, -3.21875]$

Final Approximation:

The root lies in $[-3.25, -3.21875]$. Taking the midpoint:

$$\text{Root} \approx \frac{-3.25 + (-3.21875)}{2} = -3.234375$$

Question No. 2

(a) Use Newton Raphson Method to find the root of the equation:

$$x^3 - 2x - 5 = 0$$

use the initial guess of $x_o = 2.5$ **[up to 3rd iteration]**

Solution:

Define the function and its derivative:

$$f(x) = x^3 - 2x - 5$$

$$f'(x) = 3x^2 - 2$$

The Newton-Raphson formula is:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Iterations:

1. **Iteration 1 ($x_0 = 2.5$):**

$$f(2.5) = (2.5)^3 - 2(2.5) - 5 = 5.625$$

$$f'(2.5) = 3(2.5)^2 - 2 = 16.75$$

$$x_1 = 2.5 - \frac{5.625}{16.75} \approx 2.1642$$

2. **Iteration 2 ($x_1 = 2.1642$):**

$$f(2.1642) \approx (2.1642)^3 - 2(2.1642) - 5 \approx 0.8106$$

$$f'(2.1642) \approx 3(2.1642)^2 - 2 \approx 12.0514$$

$$x_2 = 2.1642 - \frac{0.8106}{12.0514} \approx 2.0970$$

3. **Iteration 3 ($x_2 = 2.0970$):**

$$f(2.0970) \approx (2.0970)^3 - 2(2.0970) - 5 \approx 0.0265$$

$$f'(2.0970) \approx 3(2.0970)^2 - 2 \approx 11.191$$

$$x_3 = 2.0970 - \frac{0.0265}{11.191} \approx 2.0946$$

Final Approximation:

After 3 iterations, the approximate root is:

$$\text{Root} \approx 2.0946$$

$$c_1 = \frac{1.3 - 0.1}{2} = 0.6$$

$$x_1 = \frac{1.3 - 0.1}{2}\left(-\frac{1}{\sqrt{3}}\right) + \frac{1.3 + 0.1}{2} = 0.35359$$

$$c_2 = \frac{1.3 - 0.1}{2} = 0.6$$

$$x_2 = \frac{1.3 - 0.1}{2}\left(\frac{1}{\sqrt{3}}\right) + \frac{1.3 + 0.1}{2} = 1.0461$$

Then

$$\begin{aligned}\int_{0.1}^{1.3} f(x)\,dx &\approx c_1 f(x_1) + c_2 f(x_2)\\ &= 0.6f(0.35359) + 0.6f(1.04641)\end{aligned}$$

Finding the values of the function at the two quadrature points

$$\begin{aligned}f(0.35359) &= 5(0.35359)e^{-2(0.35359)}\\ &= 0.87166\end{aligned}$$

$$\begin{aligned}f(1.04641) &= 5(1.04641)e^{-2(1.04641)}\\ &= 0.64531\end{aligned}$$

we get

$$\begin{aligned}\int_{0.1}^{1.3} f(x)dx &\approx 0.6 \times 0.87166 + 0.6 \times 0.64531\\ &= 0.91018\end{aligned}$$

b) The exact value of the above integral can be found as

$$\int_{0.1}^{1.3} 5xe^{-2x}\,dx = 0.89386$$

The true error

$$\begin{aligned}E_t &= \text{True Value} - \text{Approximate Value}\\ &= 0.89386 \; - \; 0.91018\\ &= -0.01632\end{aligned}$$

c) The absolute relative true error $|\varepsilon_t|$ is

$$\begin{aligned}|\varepsilon_t| &= \left|\frac{\text{True Value} - \text{Approximate Value}}{\text{True Value}}\right| \times 100\\ &= \left|\frac{0.89386 \; - 0.91018}{0.89386}\right| \times 100\\ &= 1.8258\%\end{aligned}$$

Example 7.5.2.2

The following integral is given

$$\int_{0.1}^{1.3} 5xe^{-2x}\,dx$$

a) Use the one-point Gaussian quadrature rule to estimate the value of the integral.

b) Find the true error for part (a).

c) Find the absolute relative true error, $|\epsilon_t|$, for part (a).

Solution

a)